Questions by topic: Specialist techniques & statistical analysis - Set 1 (80 marks)

Answer all questions in the spaces provided.

0 1 Table 1 lists some techniques used in ecological monitoring.

0 1 . 1 Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation	
Image databases of unique markings			
eDNA sampling			
Geolocator tags			

0 1 . 2 Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

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0 1 . 3 Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

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0 2 Table 1 shows the mean dry mass of wheat plants grown with two fertiliser treatments (10 replicates each).

Table 1

Treatment	Mean dry mass / g	Standard deviation / g
Organic manure	14.2	3.9
Inorganic NPK fertiliser	16.1	4.1

0 2 . 1 Explain what the standard deviation values tell you about the data.

[2 marks]

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0 2 . 2 Name the statistical test that should be used to decide whether the difference between the two means is significant, and justify your choice.

[2 marks]

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0 2 . 3 The test gave a probability p = 0.30. State what this means and what the student should conclude.

[2 marks]

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0 2 . 4 Explain why a statistically significant correlation does not prove a causal relationship.

[2 marks]

0 2 . 5 Suggest how the student could increase the chance of detecting a real difference if one exists.

[2 marks]

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0 3 . 1 For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases

Whether the numbers of dormice differ between woodlands with different management

Whether mean dissolved oxygen differs between two streams

Whether crop yield increases with fertiliser application (ranked data)

0 3 . 2 Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

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0 3 . 3 A Spearman's rank test on 12 pairs of data gave $r_s = -0.72$. The critical value at $p = 0.05$ for $n = 12$ is 0.59. Explain what can be concluded.

[3 marks]

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0 3 . 4 Explain the purpose of error bars on a graph of mean values.

[2 marks]

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0 4 . 1 Describe how DNA databases are used to control the illegal trade in tropical timber and ivory.

[3 marks]

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0 4 . 2 Explain the difference between radio / GPS tracking and acoustic tracking and state when each is used.

[2 marks]

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0 4 . 3 Give **three** types of carrier system used to move sensors to study areas and one advantage of drones for ecological monitoring.

[3 marks]

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0 4 . 4 Explain how motion-sensitive cameras and CCTV reduce disturbance to wildlife compared with direct observation.

[2 marks]

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0 5 A student sampled the ground-flora of two woodlands. Table 1 shows the results for woodland A.

Table 1

Species	Number of individuals (n)
Bluebell	40
Wood anemone	30
Dog's mercury	20
Bramble	10
Total (N)	100

0 5 . 1 Use the formula $D = N(N - 1) / \Sigma n(n - 1)$ to calculate Simpson's index of diversity for woodland A.

[3 marks]

Answer

0 5 . 2 Woodland B had an index of 1.6. State what this shows and give **one** limitation of using the index alone to compare the woodlands.

[2 marks]

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0 6 A student measured the wind speed at 10 sites inside a woodland and at 10 sites in a nearby clearing.

Table 1

Location	Mean wind speed / m s^{-1}	Standard deviation / m s^{-1}
Woodland	1.2	0.4
Clearing	3.5	1.1

0 6 . 1 Name the statistical test the student should use to find out whether the difference between the means is significant. Give **two** reasons for your choice.

[3 marks]

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0 6 . 2 Explain what the student should do if it is not possible to show that the data are normally distributed.

[1 mark]

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0 6 . 3 The test gave p = 0.02. Explain what this means.

[2 marks]

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0 6 . 4 Explain how the standard deviations in Table 1 support the conclusion that the means are different.

[2 marks]

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0 6 . 5 Explain why a statistically significant result does not prove that trees cause the lower wind speed.

[2 marks]

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0 7 . 1 Complete Table 1 by naming the most suitable statistical test for each hypothesis.

[4 marks]

Table 1

Hypothesis	Statistical test	
Noise levels decrease as distance from a road increases		
Fewer cars use a road after road charging is introduced (counts on 12 days before and after)		
Woodlands managed in different ways have different numbers of dormice		
Mean lobster mass is higher in areas where collection is banned		

0 7 . 2 Explain why the chi-squared test cannot be used to compare the dissolved oxygen concentrations of two rivers.

[2 marks]

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0 7 . 3 Explain why a scientific investigation never 'proves' a hypothesis.

[2 marks]

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0 7 . 4 Describe how error bars are added to a graph of means and how they are interpreted.

[2 marks]

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0 8 Table 1 shows three investigations carried out by a class.

Table 1

Investigation	Data collected
A	Mean height of bluebells in a coppiced and an uncoppiced wood (both normally distributed)
B	Number of limpets in each of five zones on a rocky shore, compared with the numbers expected if they were evenly spread
C	Soil moisture and the percentage cover of a moss, measured at 14 points along a transect

0 8 . 1 Name the most suitable statistical test for each investigation in Table 1.

[3 marks]

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0 8 . 2 Explain what it means if a test gives a probability of $p = 0.02$.

[2 marks]

0 8 . 3 Explain why the calculated value from a test must be compared with a critical value, and what else is needed to find that critical value.

[3 marks]

0 8 . 4 Explain what the **standard deviation** of each set of bluebell heights tells the student, and how it is shown on a graph of the means.

[3 marks]

0 8 . 5 The class found a significant correlation in investigation C. Explain **four** reasons why this does not show that soil moisture causes the moss to grow.

[4 marks]

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END OF QUESTIONS

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